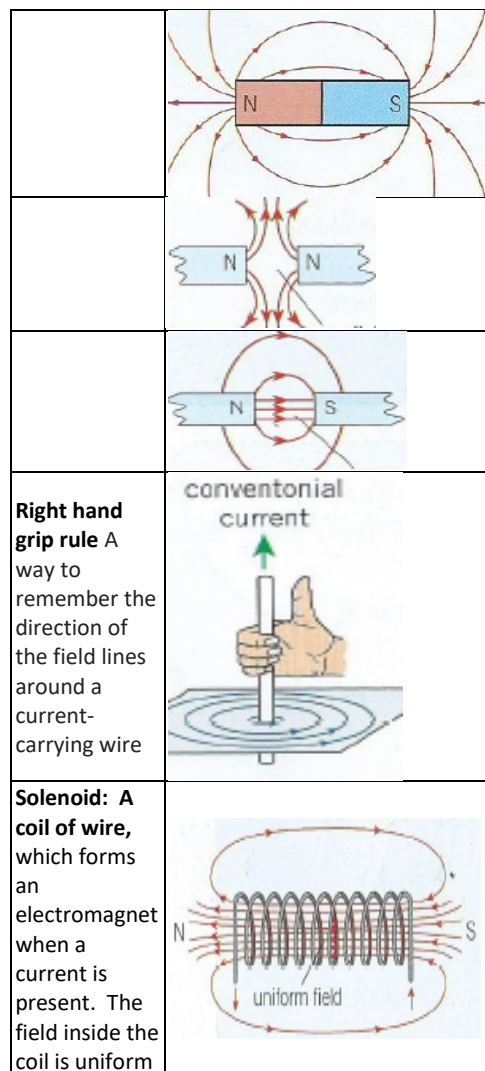


1. Magnetic fields

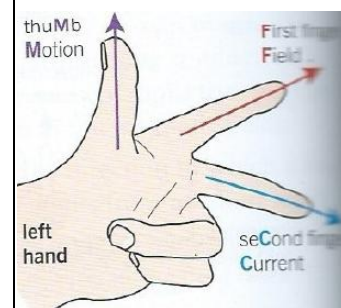
Definition	A magnetic field is a region of space in which the magnetic force can be experienced The force is experienced by magnetic materials and moving charged objects
Magnetic materials	Magnetic materials can become magnets, eg by putting them in contact with another magnet Iron, cobalt, nickel. Alloys of these (eg steel, an alloy of iron) (other materials are affected by magnets but are not considered fully 'magnetic')
Hard/soft	. A hard magnetic material will keep its magnetism well; a soft magnetic material will lose it easily (become demagnetised)
Poles	Like poles repel; opposites attract
Magnetic field lines	Lines point from the north pole of a magnet to the south. They show the direction the north pole of a plotting compass would point.
North pole	The magnetic pole at the North pole of the Earth is a south magnetic pole. We know this because the north pole of a magnet is the one which points north. Avoid confusion with the phrase 'magnetic north' from geography!
Uniform field	Straight, evenly spaced field lines represent a uniform field. The closer the lines, the stronger the field



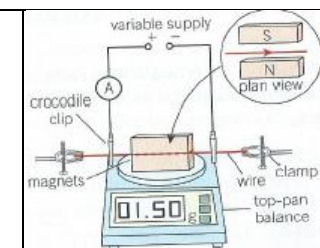
All diagrams are from '*A level Physics for OCR*' and further details can be found there.

2 Understanding magnetic field (the effect on a current-carrying wire)

Magnetic flux density, B	The strength of the magnetic field. Unit T (tesla). $1\text{T} = 1\text{Nm}^{-1}\text{A}^{-1}$ Also, $1\text{T} = 1\text{Wbm}^{-2}$, see magnetic flux, below.
Magnitude	The magnitude of the force on a current-carrying wire is given by $F = BIL$ Or $F = BIL\sin\theta$ where θ is the angle between the field lines and the current direction
Direction	Always at right angles to the plane containing the field lines and the current. If B and I are at right angles, F is at right angles to both. To work out which of the two possible directions, use Fleming's left hand rule: First finger = field Second finger = current Thumb = Force (think 'motion')



Experiment



Note the balance gives a reading of mass, which must be converted to force using $F = mg$... the mass is NOT changing, we are using the balance to measure the force.

3. Charged particles in a magnetic field

Magnitude of the force	$F = BQv$ Often the charge Q is e
Direction of the force	Always at right angles to the plane containing the field lines and the direction of motion of the particle. Fleming's left hand rule applies – think of the moving charge as a 'current' and remember that electron flow is the opposite direction to conventional current
Charged particles can be deflected into circular paths	If you covered the cyclotron at GCSE, this is how it works. See Ch16 Circular Motion for 'centripetal force' $F = BQv = \frac{mv^2}{r}$ Rearranging gives the radius of the circle: $r = \frac{mv}{BQ}$

3. Charged particles in a magnetic field (examples)

Velocity selector	Used to 'filter' a beam of particles with a range of speeds; only those with velocity v can pass through, where $v = \frac{E}{B}$ An electric (E) and magnetic (B) field pull the particles in opposite directions.
Mass spectrometer	A device used in chemical analysis, it takes a beam of particles of different masses and sorts them into order of mass. Requires a velocity selector and a uniform magnetic field to bend the particles into circular paths; the equation for the radius, above, shows that the more massive particles will turn through a larger radius
Hall effect	A Hall probe is a device to measure B. When placed in a magnetic field a pd from one side to the other develops, V_H . $V_H = \frac{BI}{nte}$ Number density n , thickness t refer to the material of the Hall probe; a thin slice of semiconductor (low n) is best.

4. Electromagnetic induction

Definition part 1	Electromagnetic induction is when an emf is induced (caused) in a wire by a moving or changing magnetic field. The magnetic field could be increasing or decreasing, or the wire or magnet could be moving. Definitions usually refer to a 'changing magnetic field' which can be misleading
	Defined as $\phi = BA \cos \theta$ Unit: Wb (weber). B is magnetic flux density, A is the area of a loop, θ is the angle shown below. When $\theta = 0$, $\phi = BA$ $1 \text{ Wb} = 1 \text{ Tm}^2$, and $1 \text{ T} = 1 \text{ Wbm}^{-2}$
Magnetic flux, ϕ	
Magnetic flux linkage	This quantity does not have its own symbol: Magnetic flux linkage = $N\phi$ N is the number of turns in a coil Unit Wb, since N has no unit
Definition part 2	With an understanding on flux linkage, we can define electromagnetic induction more briefly: "An emf is induced in a conductor when the flux linkage changes"

5. Faraday's Law, Lenz's law

Faraday's law	In electromagnetic induction, this tells us the magnitude of the emf induced $\epsilon = - \frac{\Delta(N\phi)}{\Delta t}$ The emf is equal to the rate of change of flux linkage
Lenz's law	The direction of the induced emf is such as to oppose the change causing it. The minus sign in the equation above indicates this
AC generator	Mains electricity is 230V ac, 50Hz

6. Transformers

Step-up, step-down	Step-up transformers increase the voltage (and so reduce the current). Step-down do the opposite. $\frac{n_s}{n_p} = \frac{V_s}{V_p}$
Transformer equation	P is primary coil S is secondary coil V is pd n is number of turns
100% efficiency	A 100% efficient transformer would transfer all the power to the secondary: $V_s I_s = V_p I_p$ Real transformers are highly efficient, but never 100%

Transformer core	Purpose: to transfer the magnetic flux from the primary to the secondary. Made of soft iron, laminated (in layers). Soft iron can be magnetised and demagnetised easily; the laminations prevent eddy currents (these are electrons flowing in circles within the core; if these happen, energy is lost and the core heats up.)
National grid	Step-up and step-down transformers are used to change the voltage of the electrical energy transmitted around the country. Low voltages are safer; high voltages are more efficient
Transformers	Are not robots in disguise

